

QUESTION ONE**(12 MARKS)**

- (a) Factorise completely $y^3 - 3y^2 + y - 3$. 2
- (b) Solve $x^2 - 2x - 3 \geq 0$. 2
- (c) Find a and b if $(4 - 2\sqrt{7})^2 = a + b\sqrt{7}$. 2
- (d) Solve $|2x - 1| = 9$. 2
- (e) Evaluate $\log_3 7$ correct to 3 significant figures. 2
- (f) If the line $kx + 3y - 7 = 0$ has a gradient of 2, find the value of k . 2

END OF QUESTION ONE

QUESTION TWO - Start a new booklet.**(12 MARKS)**

- (a) Solve the following simultaneous equations: 2

$$3x - y = 8$$

$$2x + 3y = -2$$

- (b) Differentiate the following:

(i) $y = (5x^3 - 4)^5$. 2

(ii) $y = \frac{2x}{x-3}$. 2

(iii) $f(x) = 3xe^{2x}$. 2

(iv) $f(x) = \ln(3x^2 + 3x)$, giving your answer in simplest form. 2

- (c) Find the exact value of $\sin 240^\circ \times \sec 45^\circ$. 2

END OF QUESTION TWO

QUESTION THREE- Start a new booklet.**(12 MARKS)**

(a) Find $\int \frac{x}{x^2 + 4} dx$. 2

(b) Evaluate $\int_0^{\frac{\pi}{6}} \sec^2 2x \, dx$, giving your answer in exact form. 3

(c) Evaluate $\int_0^2 2x e^{3x^2} dx$, giving your answer in exact form. 2

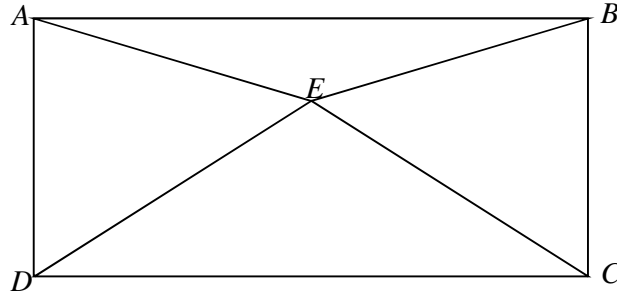
(d) Find k such that $\int_0^k (3 + 2x) dx = 4$. 3

(e) Evaluate $\sum_{r=1}^{10} 2^r$. 2

END OF QUESTION THREE

QUESTION FOUR- Start a new booklet.**(12 MARKS)**

- (a) In the rectangle $ABCD$ below, $AE = EB$ and $\angle BAE = 15^\circ$.



Copy and label the diagram above into your booklet.

- | | | |
|-------|---|---|
| (i) | Explain why $\angle DAE = \angle CBE$. | 1 |
| (ii) | Prove $\triangle DAE \equiv \triangle CBE$. | 2 |
| (iii) | Hence prove that $\triangle DEC$ is isosceles. | 1 |
| | | |
| (b) | (i) Express $y^2 - 2x - 8 = 0$, in the form $(y - k)^2 = 4a(x - h)$,
and hence find: | 1 |
| | (ii) the focal length. | 1 |
| | (iii) the co-ordinates of the vertex. | 1 |
| | (iv) the co-ordinates of the focus. | 1 |
| | (v) the equation of the directrix. | 1 |
| | (vi) the equation of the focal chord which also passes through
the point $(4, 4)$. | 3 |

END OF QUESTION FOUR

QUESTION FIVE - Start a new booklet.**(12 MARKS)**

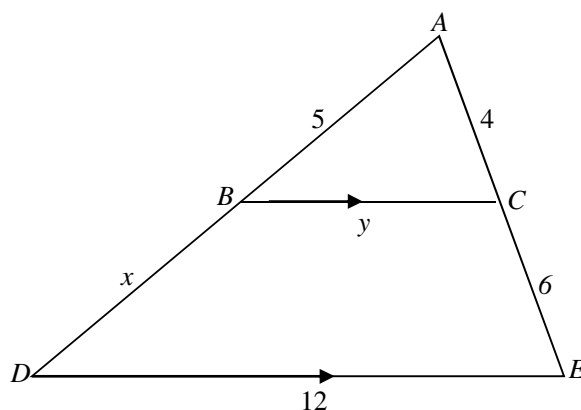
- (a) A , B , and C are points $(3, 2)$, $(-2, 4)$, and $(5, -1)$ respectively.
- (i) Draw a diagram in your booklet, representing A , B and C . 1
- (ii) Find the gradient of BC . 2
- (iii) Find the equation of BC in general form. 2
- (iv) Find the perpendicular distance of A to BC , giving your answer in simplest surd form. 2
- (b) Show that $\frac{\tan \theta \sec \theta}{1 + \tan^2 \theta} = \sin \theta$. 2
- (c) (i) Find the locus of the point $P(x, y)$, which moves so that the line PA is perpendicular to the line PB where A is $(4, 3)$ and B is $(-2, -1)$. 2
- (ii) Interpret this locus geometrically. 1

END OF QUESTION FIVE

QUESTION SIX - Start a new booklet.**(12 MARKS)**

- (a) (i) On the same axes, sketch the graphs of $y = 3x^2 + 2$ and $y = 7x$. 1
- (ii) Find the x co-ordinates of the points of intersection. 2
- (iii) Hence, use the graphs to solve the inequation $7x \geq 3x^2 + 2$. 1
- (b) Sketch the region simultaneously defined by $y \geq |x - 3|$ and $y < 3$, 2
- labelling all intercepts, and points of intersection.

(c)



Not to scale

- (i) Show that $\triangle ABC \parallel \triangle ADE$. 2
- (ii) Hence find the values of x and y . 4

END OF QUESTION SIX

QUESTION SEVEN - Start a new booklet.**(12 MARKS)**

- (a) Given that $y = 2x \log_e x$, find, giving your answers in exact form:
- (i) $\frac{dy}{dx}$. 1
 - (ii) the co-ordinates of the stationary point, and determine its nature. 4
- (b) Ayla is saving for a holiday. In the first month she saves \$30. In the second month she saves \$35. In each subsequent month her savings are \$5 more than the month before.
- (i) How much will she save in the 17th month? 1
 - (ii) How much money will she have saved in total by the 17th month? 2
 - (iii) Ayla needs \$2100 to pay for her plane ticket. How long will it take 2
her to save this amount?
- (c) Rory decided to donate \$3000 to charity. A year later he donated $\frac{7}{8}$ of this amount to the same charity. He continued to donate in this way each succeeding year.
- (i) In which year would Rory first make a donation of less than \$1000? 1
 - (ii) What is the greatest total amount that the charity can expect to receive? 1

END OF QUESTION SEVEN

QUESTION EIGHT - Start a new booklet.**(12 MARKS)**

- (a) Use Simpson's Rule with 3 function values to find an approximation for 3

$$\int_1^3 3 \log_e x \, dx, \text{ giving your answer correct to 2 decimal places.}$$

- (b) Solve $3 \tan 2x = \sqrt{3}$, $0 \leq x \leq 2\pi$. 3

- (c) A particle is travelling in a straight line, starting from rest at the origin, such that

$$\frac{d^2x}{dt^2} = \frac{8}{(t+1)^2}, \text{ where } x \text{ is displacement in metres and } t \text{ is time in seconds.}$$

- (i) Show that the acceleration is always positive. 1

- (ii) Find an expression for the velocity. 2

- (iii) Show that the distance covered between $t = 2$ seconds and $t = 5$ seconds is $(24 - 8 \log_e 2)$ metres. 3

END OF QUESTION EIGHT

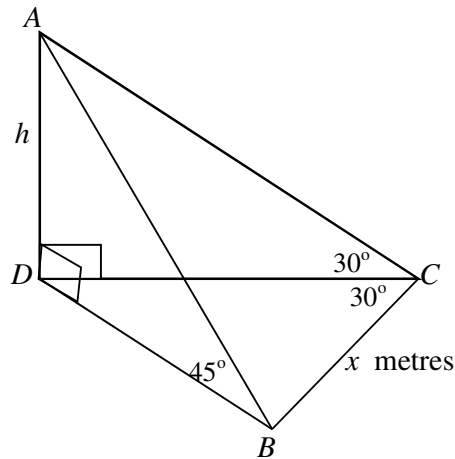
QUESTION NINE - Start a new booklet.**(12 MARKS)**

- (a) A quadratic function is given by $f(x) = x^2 + mx - 3$, and $f(-1) = 5$.
- (i) Find the value of m . 1
- (ii) If the roots of the function are α and β , find the value of $\alpha^2 + \beta^2$. 2
- (b) Calculate the volume of solid of revolution if the hyperbola $y = \frac{4}{x}$ is 2
rotated about the x axis from $x = 1$ to $x = 4$.
- (c) The number of bacteria in a culture is growing according to the formula
 $N = 300e^{kt}$, where t is time in hours.
- (i) What was the initial number of bacteria in the culture? 1
- (ii) If, after 8 hours, the number of bacteria has doubled, 2
calculate the value of k correct to 4 decimal places.
- (iii) How long would it take for the number of bacteria to rise to 3000? 2
Give your answer correct to the nearest minute.
- (iv) How many bacteria would there be after 16 hours? 1
- (v) At what rate will the bacteria be increasing after 16 hours? 1

END OF QUESTION NINE

QUESTION TEN - Start a new booklet.**(12 MARKS)**

- (a) AD , an upright tower of height h metres, is standing on level ground. The angles of elevation to the top of the tower from points C and B on the ground, are 30° and 45° respectively, and $\angle DCB = 30^\circ$.

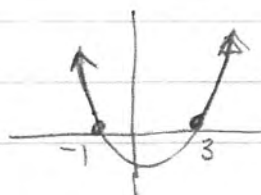


- (i) Find the lengths of BD and CD in terms of h . 2
- (ii) By considering $\triangle DCB$, show that $2h^2 = 3xh - x^2$. 3
- (b) A sector AOB , with radius R and $\angle AOB = \theta$, has an area of 625cm^2 .
- (i) Show that $\theta = \frac{1250}{R^2}$. 1
- (ii) Show that the perimeter (P) of the sector can be given 1
by:
$$P = 2R + \frac{1250}{R}.$$
- (iii) Find the value of R that is required to give a minimum perimeter. 3
- (c) Angus takes out a student loan of \$5000 at 2.25% interest per month, 2
reducible monthly. Regular repayments are made at monthly intervals.
What should each repayment be if he wishes to pay off the loan in 5 years?

END OF EXAMINATION

Q1 (a) $y^3 - 3y^2 + y - 3 = y^2(y-3) + 1(y-3)$ ①
 $= (y^2+1)(y-3)$ ①

(b) $x^2 - 2x - 3 \geq 0$
 $(x+1)(x-3) \geq 0$ ①



$x \geq 3$ or $x \leq -1$ ①

(c) $(4-2\sqrt{7})(4-2\sqrt{7}) = 16 - 16\sqrt{7} + 28$
 $\therefore a+b\sqrt{7} = 44 - 16\sqrt{7}$ ①

$a = 44, b = -16$ ①

(d) $|2x-1| = 9$ $2x-1=9$ or $2x-1=-9$
 $2x=10$ $2x=-8$
 ① $x=5$ ① $x=-4$

(e) $\log_3 7 = \frac{\log_{10} 7}{\log_{10} 3} = 1.77$ (3 sf) ①

(f) $kx + 3y - 7 = 0$ $3y = -kx + 7$
 $y = \frac{-k}{3}x + \frac{7}{3}$ ①

$\frac{-k}{3} = 2$ $k = -6$ ①

$$\begin{array}{rcl} \text{Q2 (a)} & 3x - y = 8 & \text{--- (1)} \quad \times 3 \quad 9x - 3y = 24 \\ & 2x + 3y = -2 & \text{--- (2)} \end{array}$$

$$\begin{array}{rcl} \text{(1) + (2)} & 11x = 22 & \\ & \underline{x = 2} & \text{(1)} \end{array} \quad \begin{array}{rcl} & y = 3(2) - 8 & \\ & \underline{= -2} & \text{(1)} \end{array}$$

$$\begin{aligned} \text{(b) i} \quad y &= (5x^3 - 4)^5 \\ y' &= 5(5x^3 - 4)^4 (15x^2) \quad \text{(1)} \\ &= 75x^2 (5x^3 - 4)^4 \quad \text{(1)} \end{aligned}$$

$$\begin{aligned} \text{ii} \quad y &= \frac{2x}{x-3} \\ y' &= \frac{(x-3)2 - 2x(1)}{(x-3)^2} \quad \text{(1)} = \frac{2x-6-2x}{(x-3)^2} \\ &= \frac{-6}{(x-3)^2} \quad \text{(1)} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad f(x) &= 3xe^{2x} \\ f'(x) &= e^{2x} \cdot 3 + 3x \cdot 2e^{2x} \quad \text{(1)} \\ &= 3(e^{2x} + 2xe^{2x}) \\ &= \underline{3e^{2x}(1+2x)} \quad \text{(1)} \end{aligned}$$

$$\begin{aligned} \text{(iv)} \quad f(x) &= \ln(3x^2 + 3x) \\ f'(x) &= \frac{6x+3}{3x^2+3x} = \frac{\cancel{3}(2x+1)}{\cancel{3}(x^2+x)} \\ &= \frac{2x+1}{x^2+x} \end{aligned}$$

$$\text{(c)} \quad \approx 240 \times \sec 45 = \underbrace{-\frac{\sqrt{3}}{2}}_{\text{(1)}} \times \sqrt{2} = \underline{-\frac{\sqrt{6}}{2}} \quad \text{(1)}$$

$$Q2(a) \int \frac{x}{x^2+4} dx = \frac{1}{2} \int \frac{2x}{x^2+4} dx \quad (i)$$

$$= \frac{1}{2} \ln(x^2+4) + C \quad (1)$$

$$(b) \int_0^{\pi/6} \sec^2 2x dx = \left[\frac{\tan 2x}{2} \right]_0^{\pi/6} \quad (1)$$

$$= \frac{1}{2} \left[\tan \frac{\pi}{3} - \tan 0 \right] \quad (1)$$

$$= \frac{1}{2} [\sqrt{3} - 0] = \frac{\sqrt{3}}{2} \quad (1)$$

$$(c) \int_0^2 2x e^{3x^2} dx = \frac{1}{3} \int_0^2 6x e^{3x^2} dx \quad (i)$$

$$= \frac{1}{3} \left[e^{3x^2} \right]_0^2$$

$$= \frac{1}{3} [e^{12} - e^0]$$

$$= \frac{e^{12} - 1}{3} \quad (1)$$

$$(d) \int_0^k (3+2x) dx = 4$$

$$\left[3x + x^2 \right]_0^k = 4 \quad (1)$$

$$[(3k + k^2) - (0)] = 4$$

$$k^2 + 3k - 4 = 0$$

$$\therefore k = 1 \text{ or } -4 \quad (1) \quad (k-1)(k+4) = 0$$

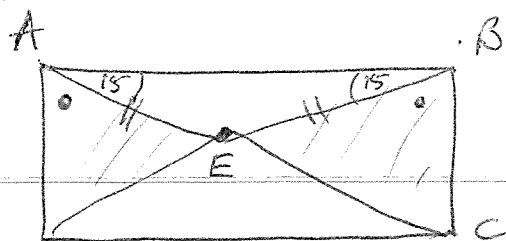
$$\text{but } k > 0 \quad \therefore \underline{\underline{k = 1}} \quad (1)$$

$$(e) \sum_{r=1}^{10} 2^r = 2 + 2^2 + 2^3 + \dots + 2^{10}. \quad (1)$$

$\therefore$ a GP with $a=2$, $r=2$.

$$S_{10} = \frac{2(2^{10}-1)}{2-1} = \underline{2046} \quad (1)$$

4(a)



D

(i) $\angle EAB = \angle EBA$ (isos Δ).

$\therefore \angle DAE = \angle CBE$ (right angles). (1)

(ii) In Δ s DAE, CBE

$AD = CB$ (opp sides rect)

$\angle DAE = \angle CBE$ (proved).

$AE = BE$ (given). (1)

$\therefore \Delta DAE \equiv \Delta CBE$ (SAS) (1)

(iii) $DE = EC$ (corr sides, cong Δ s) (1)

$\therefore \Delta DEC$ is isos.

(b) $y^2 - 2x - 8 = 0$

(i)

$y^2 = 2x + 8$

$y^2 = 2(x + 4)$ (1)

(ii) $4a = 2 \therefore a = \frac{1}{2}$ (1)

(iii) Vertex = $(-4, 0)$ (1)

(iv) Focus = $(-3\frac{1}{2}, 0)$ (1)

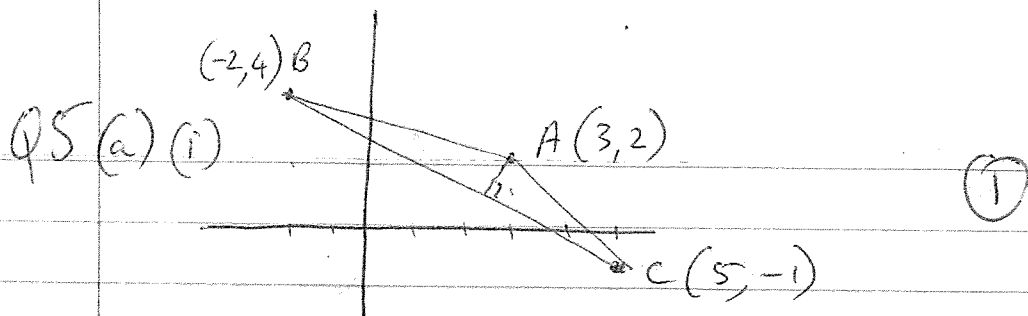
(v) Dir: $x = -4\frac{1}{2}$ (1)

(vi) grad focal chord = $\frac{4-0}{4+3\frac{1}{2}} = \frac{8}{15}$ (1)

$\therefore$ Equ: $y - 0 = \frac{8}{15}(x + 3\frac{1}{2})$ (1)

$15y = 8x + 28$

$\therefore 8x - 15y + 28 = 0$ (1)



(ii) $m(BC) = \frac{-1-4}{5-(-2)} = \frac{-5}{7}$ ①

(iii) Equ BC $y-4 = \frac{-5}{7}(x+2)$ ①

$$7y-28 = -5x-10$$

$$\underline{5x+7y-18=0} \quad \text{①}$$

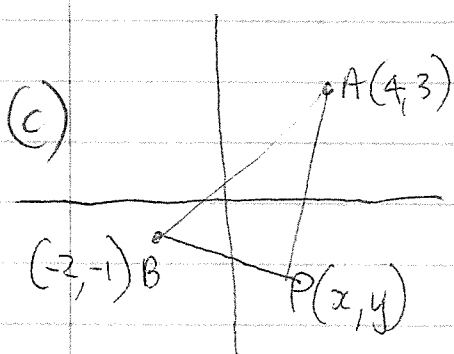
(iv) $d = \frac{|3 \times 5 + 2 \times 7 - 18|}{\sqrt{25+49}} = \frac{11}{\sqrt{74}}$ ①

$$\underline{d = \frac{11\sqrt{74}}{74}} \quad \text{①}$$

(b) $\frac{\tan \theta \sec \theta}{1+\tan^2 \theta} = \frac{\tan \theta \sec \theta}{\sec^2 \theta}$ ①

$$= \frac{\sin \theta}{\cos \theta} \div \frac{1}{\cos \theta}$$

$$= \frac{\sin \theta}{\cancel{\cos \theta}} \times \cancel{\cos \theta} = \underline{\sin \theta} \quad \text{①}$$



$$m(PA) = \frac{y-3}{x-4}$$

$$m(PB) = \frac{y+1}{x+2}$$

$$m_1 m_2 = -1$$

①

$$\frac{y-3}{x-4} \times \frac{y+1}{x+2} = -1$$

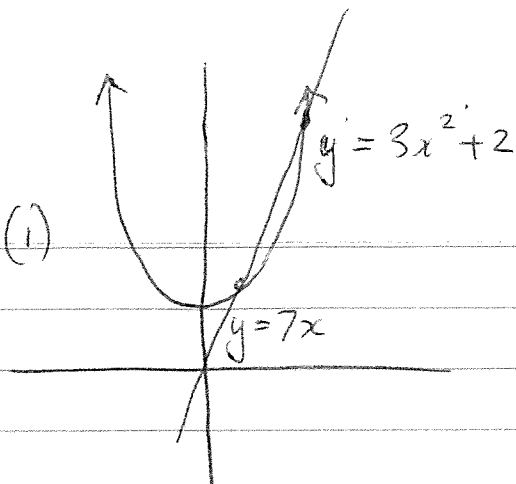
(i) $y^2 - 2y - 3 = -(x^2 - 2x - 8)$

$$y^2 - 2y + 3 = -x^2 + 2x + 8$$

$$y^2 - 2y + x^2 - 2x = 11$$

(ii) $(y-1)^2 + (x-1)^2 = 11+1+1$ ①

Q6 (a) (i)



(1)

(ii) $3x^2 + 2 = 7x$

$3x^2 - 7x + 2 = 0$

$(3x - 1)(x - 2) = 0$

$x = \frac{1}{3}, 2$

$x = \frac{1}{3}, y = \frac{7}{3} = 2\frac{1}{3}$

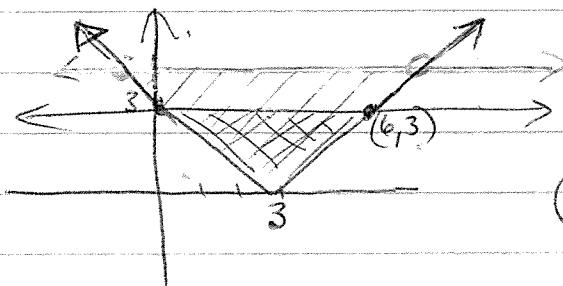
(1)

$x = 2, y = 14$

(iii) $7x \geq 3x^2 + 2$

$\frac{1}{3} \leq x \leq 2$ (1)

(b) $y \geq |x - 3|$ $y < 3$



(2)

(c) (i) In Δs ABC, ADE

$\angle ABC = \angle ADE$ (corr $\angle$'s, $\parallel$ lines)

$\angle BAC = \angle DAE$ (common) (1)

$\therefore \Delta ABC \parallel \Delta ADE$ (AA) (1)

(ii) $\frac{5}{5+x} = \frac{4}{10} = \frac{y}{12}$

$50 = 20 + 4x$ (1)

$30 = 4x$

$7\frac{1}{2} = x$ (1)

$48 = 10y$ (1)

$\frac{48}{10} = y$

$4.8 = y$ (1)

Q7(a) $y = 2x \log x$

$$(i) \frac{dy}{dx} = \log x \cdot 2 + 2x \frac{1}{x}$$

$$= 2 \log x + 2 = 2 (\log x + 1) \quad (1)$$

(ii) for SPs $\frac{dy}{dx} = 0$

$$2 (\log x + 1) = 0 \quad \therefore \log_e x = -1$$

$$e^{-1} = x \quad (1)$$

$$\therefore x = \frac{1}{e}, \quad y = \frac{2}{e} \log e^{-1}$$

$$= \frac{2}{e} \cdot -1 = -\frac{2}{e} \quad (1)$$

$$\therefore \text{SP at } \left(\frac{1}{e}, -\frac{2}{e} \right)$$

Test Nature: $\frac{d^2y}{dx^2} = \frac{2}{x} \quad (1)$

at $x = \frac{1}{e}$ $\frac{d^2y}{dx^2} = 2e > 0 \quad \therefore \text{Min TP} \quad (1)$

(b) $30 + 35 + 40 + \dots$ AP with $a = 30, d = 5$

(i) $T_{17} = 30 + (16 \times 5) = \underline{\underline{\$110}} \quad (1)$

(ii) $S_{17} = \frac{17}{2} [60 + 16 \times 5] = \underline{\underline{\$1190}} \quad (1)$

(iii) $S_n = 2100$ $2100 = \frac{n}{2} [60 + (n-1)5]$

$$4200 = n [60 + 5n - 5] \quad (1)$$

$$4200 = 55n + 5n^2$$

$$5n^2 + 55n - 4200 = 0$$

$$\begin{aligned} (\pm 5) \quad n^2 + 11n - 840 &= 0 \\ (n + 35)(n - 24) &= 0 \end{aligned}$$

$$\therefore n = -\cancel{35}, 24$$

①

$\therefore$ 24 months to get \$2100

$$(c) \quad a = 3000 \quad r = \frac{7}{8}$$

$$(i) \quad T_n < 1000$$

$$3000 \left(\frac{7}{8}\right)^{n-1} < 1000$$

$$\left(\frac{7}{8}\right)^{n-1} < \frac{1}{3}$$

$$\text{Take logs} \quad (n-1) \log\left(\frac{7}{8}\right) < \log\left(\frac{1}{3}\right)$$

$$n-1 < \frac{\log \frac{1}{3}}{\log \frac{7}{8}} \quad \swarrow \begin{matrix} (-ve) \\ \therefore \text{change} \\ \text{inequality} \end{matrix}$$

$$n-1 > 8.22$$

$$n > 9.22$$

$\therefore$ In the 10th year. ①

$$(ii) \quad S_{\infty} = \frac{3000}{1 - \frac{7}{8}} = \underline{\underline{\$24000}} \quad ①$$

Q8(a)

x	1	2	3
y	0	$3\ln 2$	$3\ln 3$

• Simpson's Rule

$$\int_1^3 3 \ln x \, dx = \frac{1}{3} [0 + 4(3\ln 2) + 3\ln 3] \quad (1)$$

$$= \frac{1}{3} [12\ln 2 + 3\ln 3] \quad (1)$$

$$= 3.87 \text{ (2dp)} \quad (1)$$

(b) $3\ln 2x = \sqrt{3} \quad \tan 2x = \frac{\sqrt{3}}{3} = \frac{1}{\sqrt{3}}$

Let $u = 2x$
 $(0 \leq u \leq 4\pi)$ $\tan u = \frac{1}{\sqrt{3}} \quad (1)$

$$u = \frac{\pi}{6}, \frac{7\pi}{6}, \frac{13\pi}{6}, \frac{19\pi}{6} \quad (1)$$

$$\therefore x = \frac{\pi}{12}, \frac{7\pi}{12}, \frac{13\pi}{12}, \frac{19\pi}{12} \quad (1)$$

(c) $\frac{d^2x}{dt^2} = \frac{8}{(t+1)^2} = 8(t+1)^{-2}$

(i) $t > 0, (t+1)^2 > 0 \quad \therefore \frac{d^2x}{dt^2} > 0 \quad (1)$

(ii) $v = \int 8(t+1)^{-2} dt = \frac{8(t+1)^{-1}}{-1} + C \quad (1)$

When $t=0, v=0 \quad 0 = -8 + C \quad \therefore C=8$

$$\therefore v = \frac{-8}{(t+1)} + 8 \quad (1)$$

$$8c \text{ (iii)} \quad x = \int_2^5 \left(\frac{-8}{t+1} + 8 \right) dt$$

$$= -8 \int_2^5 \frac{1}{t+1} dt + \int_2^5 8 dt \quad (1)$$

$$= \left[-8 \ln(t+1) \right]_2^5 + \left[8t \right]_2^5$$

$$= \left[(-8 \ln 6 + 8 \ln 3) \right] + \left[(40 - 16) \right] \quad (1)$$

$$= \left[-8(\ln 6 - \ln 3) \right] + 24$$

$$= -8 \ln 2 + 24$$

$$= \underline{24 - 8 \ln 2} \quad (1)$$

or

$$t=2, \quad x=16-8 \ln 3$$

$$t=5, \quad x=40-8 \ln 6$$

$$(40-8 \ln 6) - (16-8 \ln 3)$$

$$= 24 - 8(\ln 6 - \ln 3)$$

$$= 24 - 8 \ln 2$$

Q9 (a) $f(x) = x^2 + mx - 3$

(i) $f(-1) = 5 \quad \therefore 5 = 1 - m - 3$

$m = -7$

(1)

(ii) $\left. \begin{array}{l} \alpha + \beta = 7 \\ \alpha\beta = -3 \end{array} \right\} \textcircled{1} \quad f(x) = x^2 - 7x - 3$

$$\begin{aligned} \alpha^2 + \beta^2 &= (\alpha + \beta)^2 - 2\alpha\beta \\ &= 49 + 6 \\ &= \underline{55} \end{aligned} \quad \textcircled{1}$$

(b) $V = \pi \int_1^4 \frac{16}{x^2} dx = \pi \int_1^4 16x^{-2} dx \quad \textcircled{1}$

$$= \pi \left[\frac{16x^{-1}}{-1} \right]_1^4 = \pi \left[-\frac{16}{x} \right]_1^4$$

$$= \pi \left[(-4) - (-16) \right] = \underline{\underline{12\pi \text{ u}^3}} \quad \textcircled{1}$$

(c)

$$N = 300e^{kt}$$

(i) $t=0, N=300 \quad \textcircled{1}$

(ii) $t=8, N=600$

$$600 = 300e^{8k} \quad \textcircled{1}$$

$$2 = e^{8k}$$

Take logs

$$\ln 2 = 8k \ln e$$

$$\frac{\ln 2}{8} = k$$

$$\therefore \underline{k = 0.0866} \quad \textcircled{1}$$

(iii) $3000 = 300e^{0.0866t}$

$$10 = e^{0.0866t} \quad \textcircled{1}$$

$$\ln 10 = 0.0866t \ln e$$

$$t = \frac{\ln 10}{0.0866} = 26.5887 \dots$$

$$= \underline{26 \text{ hrs } 35 \text{ mins}} \quad (1)$$

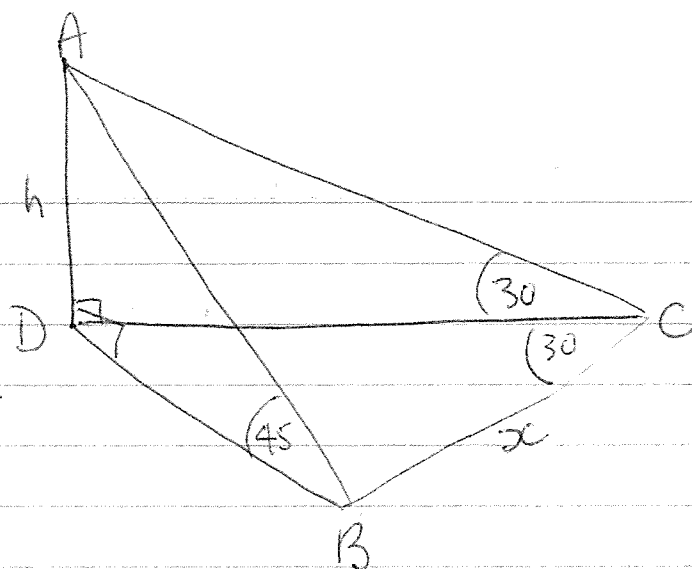
$$(iv) \quad t = 16 \quad N = 300 e^{16 \times 0.0866}$$

$$\underline{N = 1199} \quad (i)$$

$$(v) \quad \frac{dN}{dt} = KN = 0.0866 \times 1199$$

$$= \underline{103 \text{ bact/hr}} \quad (1)$$

Q10 (a)



(i) To find BD

In $\triangle ABD$

$$\tan 45 = \frac{h}{BD}$$

$$\therefore BD = \frac{h}{\tan 45} = h \quad (1)$$

To find CD

In $\triangle ACD$

$$\tan 30 = \frac{h}{DC}$$

$$\therefore DC = \frac{h}{\tan 30} = \sqrt{3}h \quad (1)$$

(ii) In $\triangle DCB$ Using Cosine Rule:

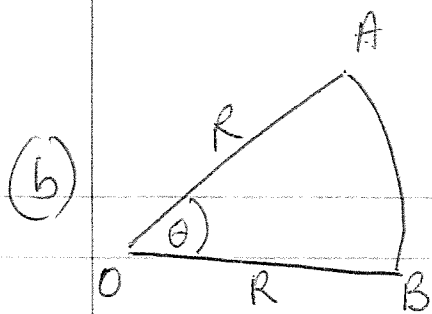
$$DB^2 = BC^2 + DC^2 - 2BC \cdot DC \cos C \quad (1)$$

$$h^2 = x^2 + 3h^2 - 2x \cdot \sqrt{3}h \cos 30$$

$$h^2 = x^2 + 3h^2 - 2\sqrt{3}xh \frac{\sqrt{3}}{2} \quad (1)$$

$$h^2 = x^2 + 3h^2 - 3xh$$

$$\therefore \underline{2h^2 = 3xh - h^2} \quad (1)$$



(i) $A = \frac{1}{2} R^2 \theta$

$$625 = \frac{1}{2} R^2 \theta$$

$$\frac{1250}{R^2} = \theta \quad (1)$$

(ii) $P = 2R + l$ and $l = R\theta$

$$P = 2R + R\theta$$

but $\theta = \frac{1250}{R^2}$

$$P = 2R + \frac{1250}{R}$$

(1)

(iii) $P = 2R + 1250R^{-1}$
 $\frac{dP}{dR} = 2 - 1250R^{-2}$

$$= 2 - \frac{1250}{R^2} \quad (1)$$

for SPs $\frac{dP}{dR} = 0$

$$\frac{1250}{R^2} = 2$$

$$R^2 = 625$$

$$R = \sqrt{625}$$

$$= 25$$

(1)

Test Nature

$$\frac{d^2P}{dR^2} = 2500R^{-3} = \frac{2500}{R^3} > 0$$

$\therefore$ MIN TP when $R = 25$ (1)

10(c) loan = \$5000 @ 2.25% p.m

$$A_n = 5000(1.0225)^{60} - \frac{m(1.0225^{60} - 1)}{0.0225} \quad (1)$$

$$A_n = 0$$

$$\therefore \frac{m(1.0225^{60} - 1)}{0.0225} = 5000(1.0225)^{60}$$

$$\therefore m = \frac{5000(1.0225)^{60} \times 0.0225}{1.0225^{60} - 1}$$

$$= \underline{\underline{\$152.68}} \quad (1)$$